Karthik P

Pathanamthitta, India

7012041942 itskp2004@gmail.com LinkedIn GitHub Portfolio

Technical Skills

- **Programming Languages:** Python, Java, SQL, C, C++, JavaScript
- **Data Engineering & Big Data:** Apache Hadoop, Apache HBase, Apache Pig, PySpark, ETL Concepts, Data Processing
- **Databases:** PostgreSQL, MySQL, Relational Database Design, SQL Queries
- **Python Libraries:** Pandas, NumPy, Statsmodels
- **Backend & APIs:** Spring Boot, FastAPI, Node.js, REST APIs
- **Tools & Technologies:** Git, GitHub, Docker, React.js, Next.js
- **AI & Machine Learning:** PyTorch, TensorFlow, Keras, OpenCV

Education

Amrita Vishwa Vidyapeetham, Coimbatore
*B.Tech Computer Science & Engineering (CGPA of **7.11**)* **2023-2027**

St. Mary's Residential School, Pathanapuram, Kollam
*Higher Secondary School, ISC, Science with Computer Science (**86%**)* **2022-2023**
*Secondary School, ICSE (**96.2%**)* **2020-2021**

Projects

NYC Yellow Taxi Data Processing | *Java, Apache HBase, Hadoop* | Academic Project

- Designed an Apache HBase table for NYC Yellow Taxi trip records, organizing transactional attributes into separate column families for trip, payment, and location information.
- Developed a Java-based data ingestion program to load structured CSV records into HBase and implemented data retrieval, filtering, and querying using HBase Shell and Java API.

Impact of Weather on Retail Sales | *Python, Pandas, Statsmodels, Chart.js* | Group Project

- Integrated 2014–2018 grocery sales data with real-world weather records, performing data cleaning, transformation, and exploratory data analysis to study the impact of temperature, rainfall, and seasonality on consumer behavior.
- Applied hypothesis testing and OLS regression to identify relationships between weather conditions and sales, presenting findings through an interactive Chart.js dashboard.

MARINA | *Java, Python* | Group Project

- Designed and implemented an intelligent vehicular edge computing system using LSTM-based mobility prediction models to anticipate vehicle movement patterns and enable proactive task scheduling.
- Optimized computational workload allocation between mobile vehicles and roadside base stations based on predicted mobility and resource availability, reducing execution cost while meeting deadline constraints.

Hybrid Stock Predictor Web App | *Python, FastAPI, FinBERT, yfinance, React* | Individual Project

- Developed a financial data analysis platform integrating live stock market data through yfinance and financial news sentiment analysis using FinBERT.
- Implemented a weighted prediction approach combining news sentiment (70%) and technical indicators (30%) when recent news is available, delivering daily BUY/SELL/HOLD signals through a FastAPI backend and React dashboard.

Secure Prescription Management Platform | *React, Spring Boot, PostgreSQL, Docker* | Individual Project

- Developed a full-stack prescription management platform using Spring Boot REST APIs and PostgreSQL for structured prescription and user data management.
- Implemented Multi-Factor Authentication (MFA) and Role-Based Access Control (RBAC), and containerized the application stack using Docker Compose for consistent deployment.

Certifications

- IBM AI Engineering Professional Certificate (13-course series covering Machine Learning, Deep Learning, Generative AI, LLMs, and RAG systems)